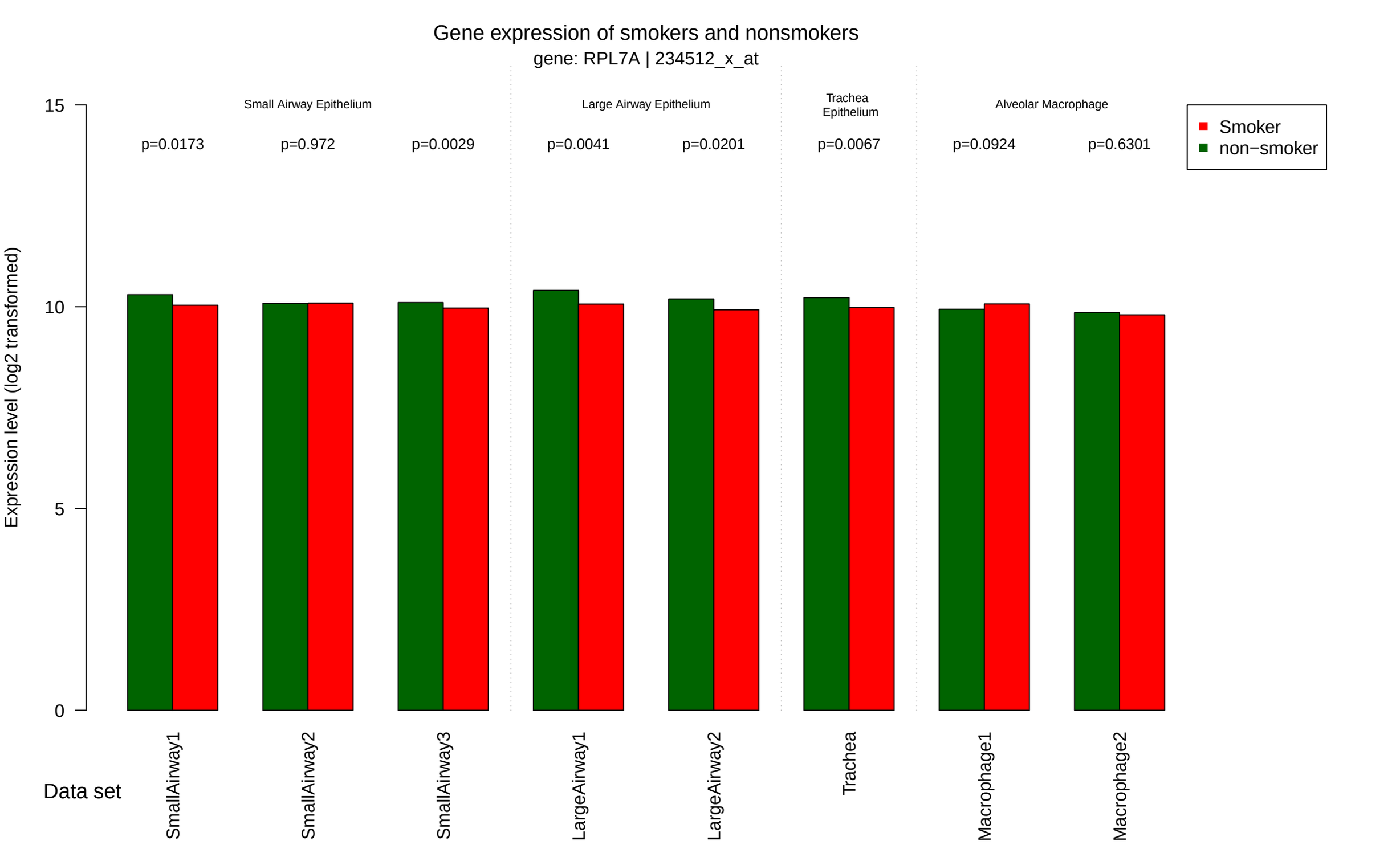
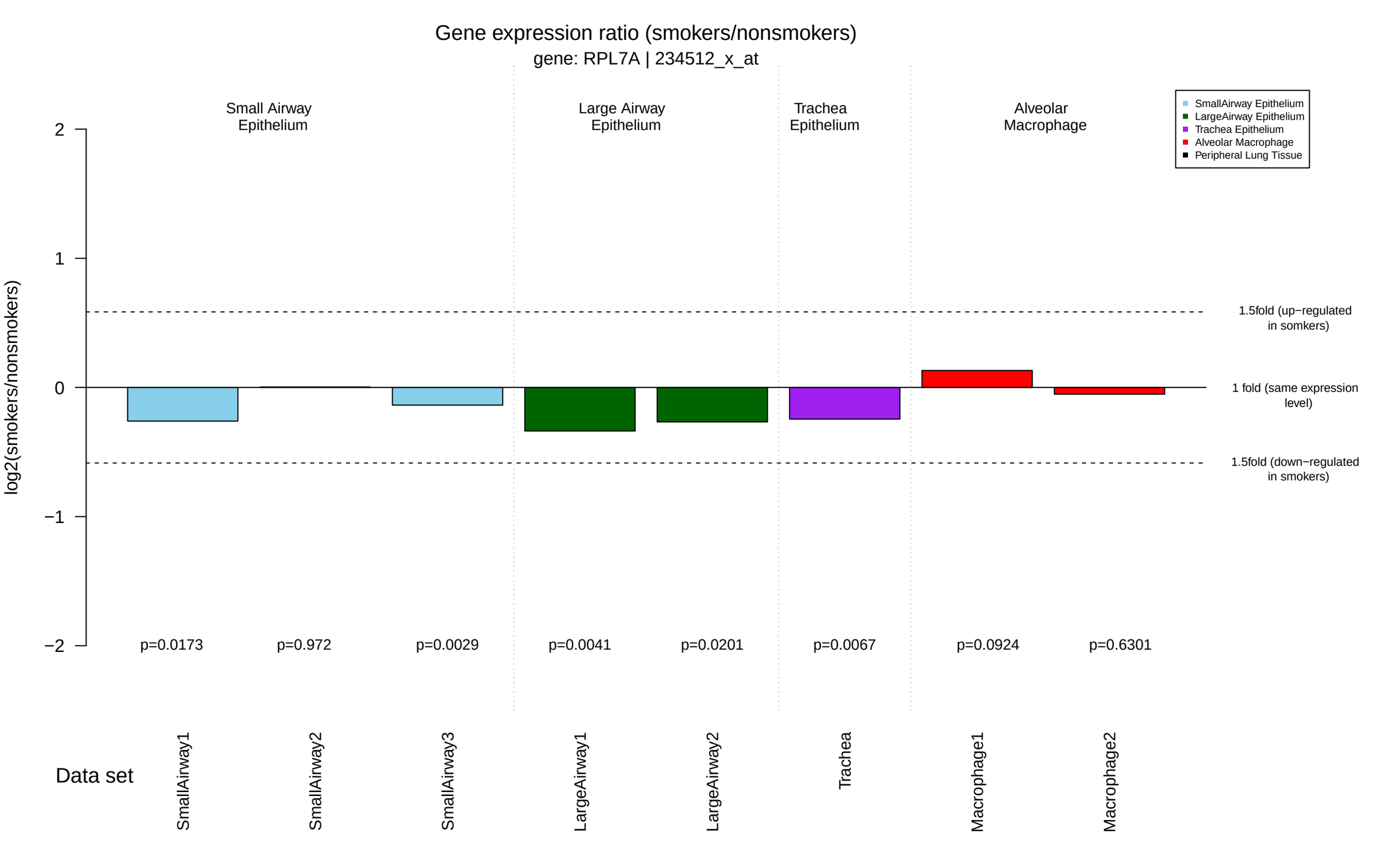
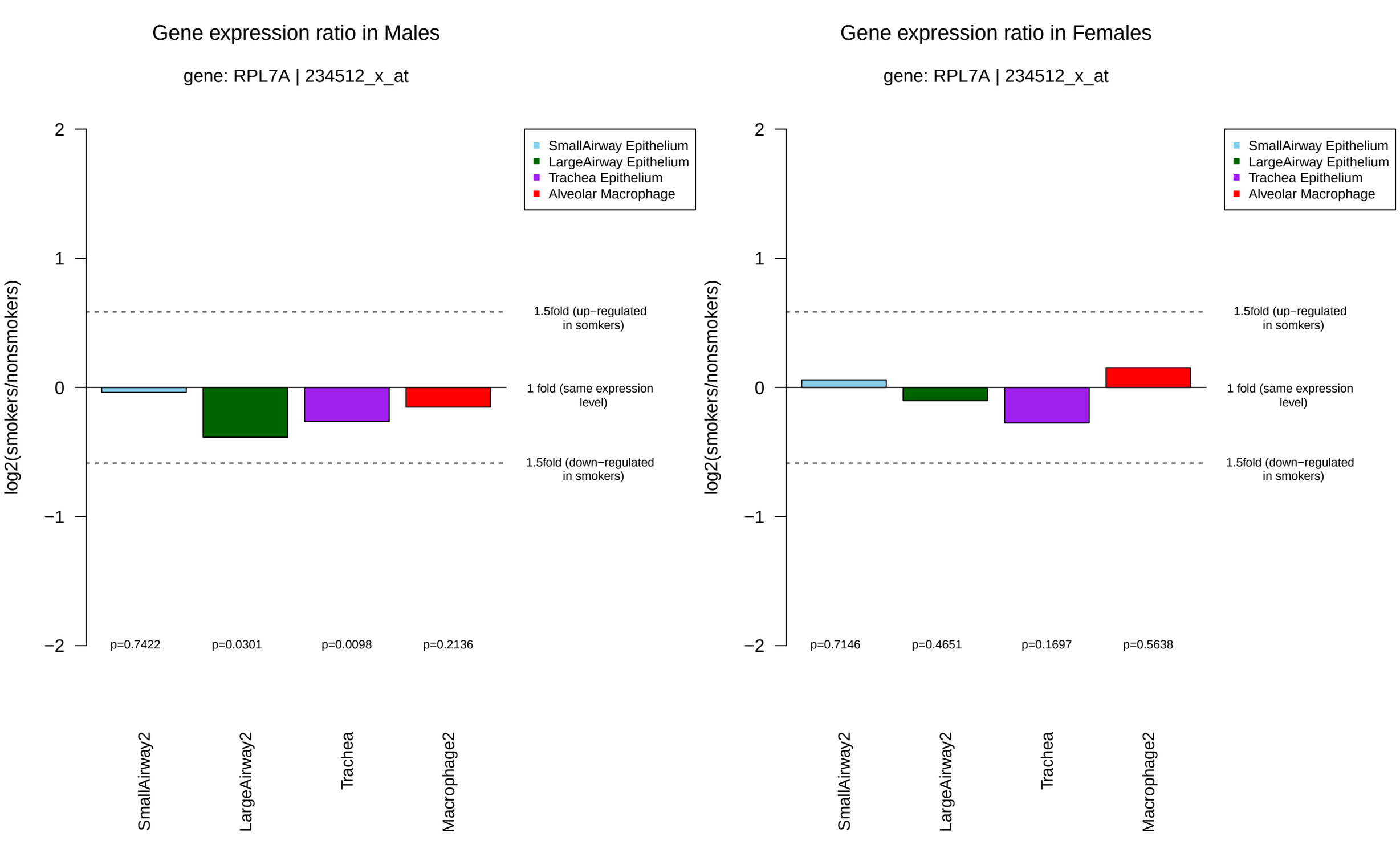
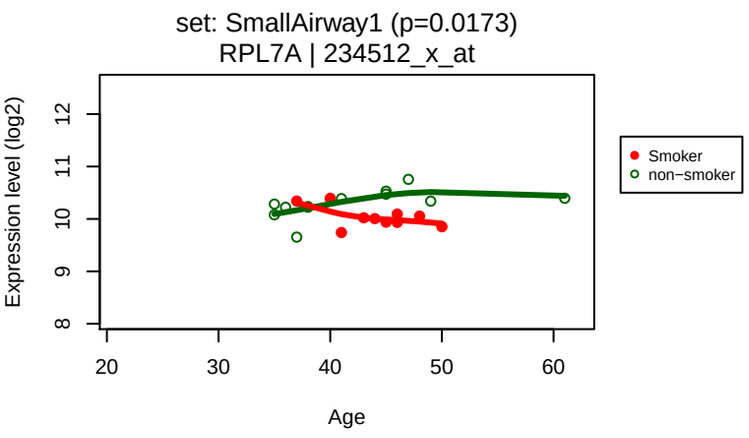
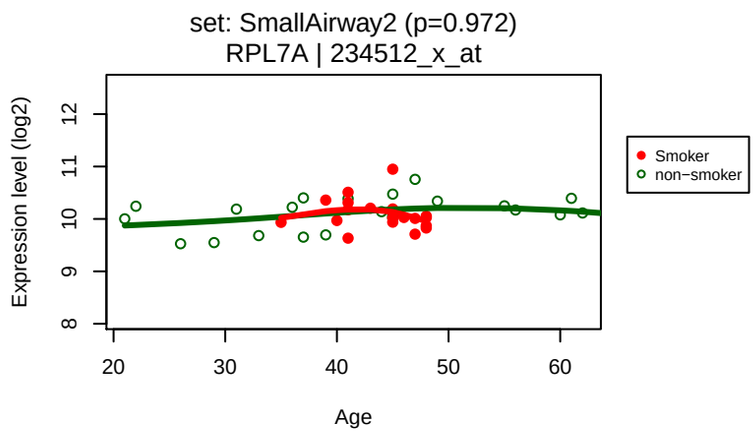
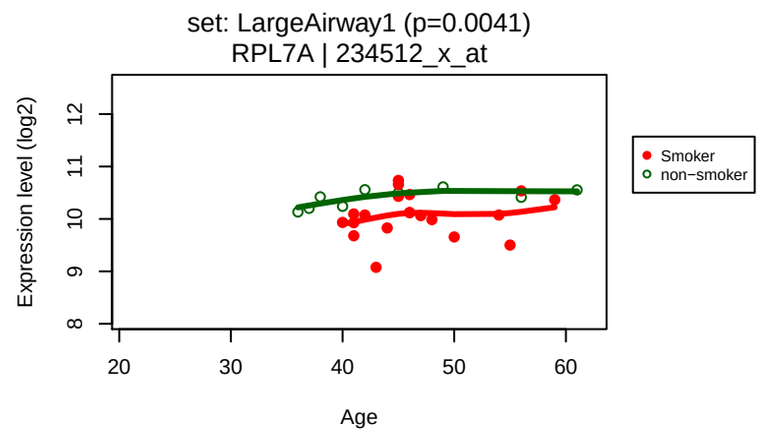

Gene expression of smokers and nonsmokers
gene: RPL7A | 234512_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0173
p=0.972
p=0.0029
p=0.0041
p=0.0201
p=0.0067
p=0.0924
p=0.6301
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

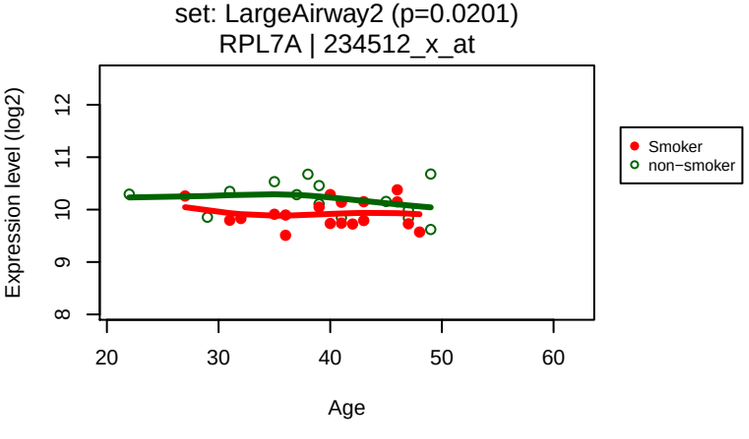

Gene expression ratio (smokers/nonsmokers)
gene: RPL7A | 234512_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.972
p=0.0029
p=0.0041
p=0.0201
p=0.0067
p=0.0924
p=0.6301
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

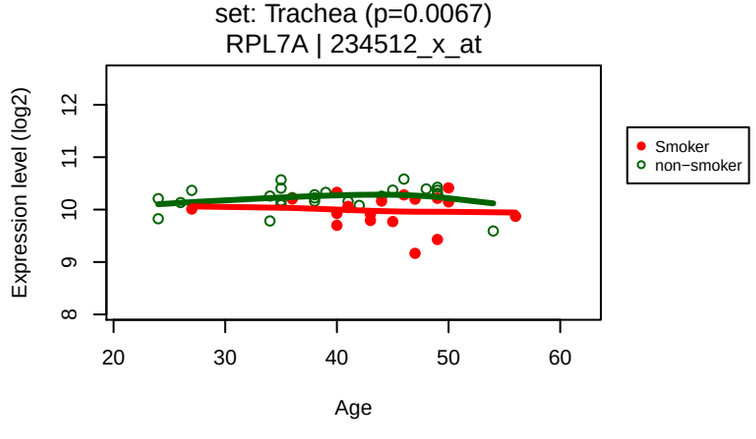

Gene expression ratio in Males
gene: RPL7A | 234512_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7422
p=0.0301
p=0.0098
p=0.2136
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL7A | 234512_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7146
p=0.4651
p=0.1697
p=0.5638
SmallAirway2
LargeAirway2
Trachea
Macrophage2

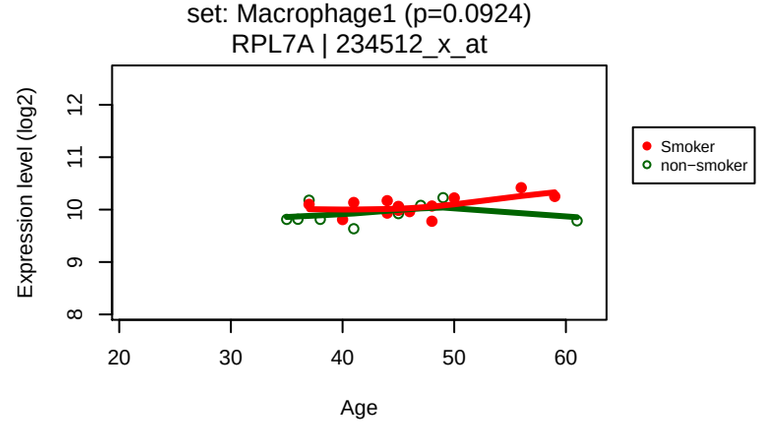

set: SmallAirway1 (p=0.0173)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker

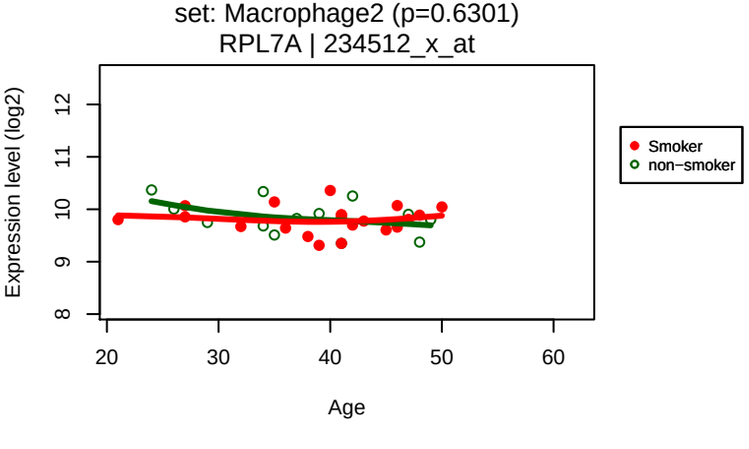

set: SmallAirway2 (p=0.972)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker

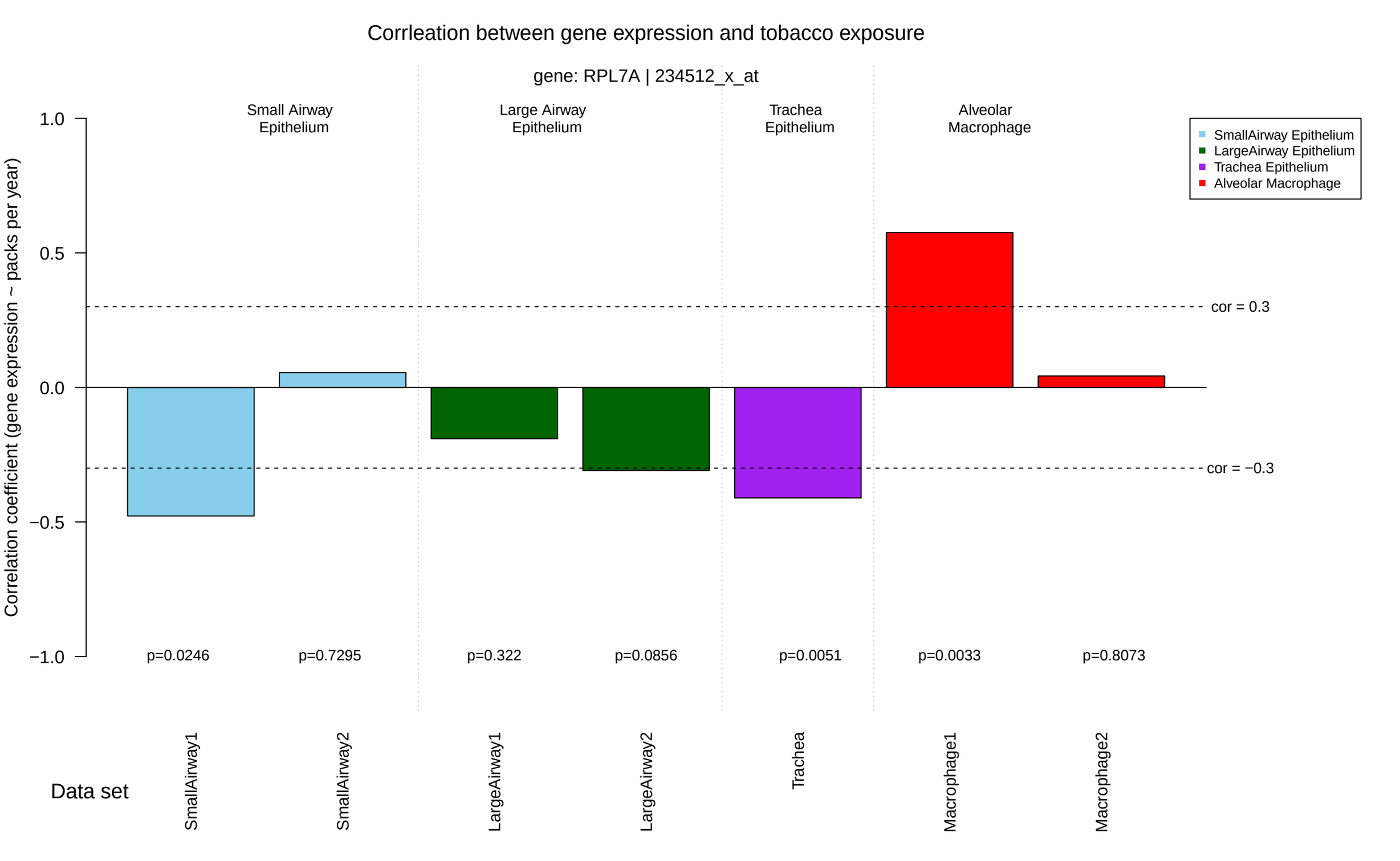

set: LargeAirway1 (p=0.0041)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0201)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0067)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0924)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6301)
RPL7A | 234512_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL7A | 234512_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0246
p=0.7295
p=0.322
p=0.0856
p=0.0051
p=0.0033
p=0.8073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2